BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 107 · Lesson 107 · Paging, Segmentation and Virtual Memory

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Identify two services provided by an OS.

Q2. What is the time complexity of sequential search in the worst case?

Q3. After pushing A then B onto a stack, which is popped first?

Q4. Why must a binary search operate on ordered data?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

page — Fixed-size virtual-memory block.

frame — Fixed-size physical-RAM allocation unit.

segmentation — Memory organised into variable-size logical segments.

page fault — Access to a page not currently resident in RAM.

New learning

Paging — Virtual memory is split into fixed-size pages; RAM into frames.

Segmentation — Logical variable-size segments represent code, data or stack regions.

Virtual memory — The OS maps addresses and can move inactive pages to secondary storage.

Worked example

A 2,500-byte virtual address uses 1,024-byte pages. Find the page and offset, then place it in frame 7.

1. Model answer — Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7. — Check: link each claim to the scenario and use precise OCR terminology.

We Do · Practise with guidance

A 4,100-byte virtual address uses 1,024-byte pages; page 4 maps to frame 1. Find the physical address.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

4,100 ÷ 1,024 gives page 4 and offset 4. Frame 1 starts at 1,024 so physical address = 1,028.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Differentiate paging and segmentation, with one reason for using each. [6]

Question 2

With 4,096-byte pages, calculate page number and offset of virtual address 8,193. [4]

Question 3

Explain why heavy paging can cause thrashing and slow down a computer. [8]

Exit Ticket · Answer before checking

Virtual memory increases the speed of every memory access.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Any two: process management, memory management, files, devices, security or interface.

Q2. O(n).

Q3. B, because a stack is LIFO.

Q4. It discards half the search space using an ordering comparison.

You Do pages · Independent questions

Q1. Paging uses fixed-size blocks [2], segmentation variable-size logical blocks [2], and both support memory mapping/protection/management [2].

Q2. Page = 2 [2]; offset = 1 [2], as 2×4,096 = 8,192.

Q3. Too few resident pages [2]; page faults increase [2]; frequent disk/SSD transfers are slower than RAM [2]; reduced useful CPU execution and mitigation via more RAM/reduced working set [2].

Exit Ticket · Model answer

False. Page faults can require slow secondary-storage access.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.